

RailVision -- Strategic Overview

Executive Summary

RailVision is an AI-powered, locomotive-agnostic technology platform purpose-built for the North American shortline railroad market -- a segment largely ignored by incumbent enterprise rail technology providers.

With approximately 15,000 shortline locomotives in service across North America, the vast majority operating on aging Tier 0 diesel equipment with no access to modern fuel optimization tools, RailVision has identified a greenfield opportunity to deliver measurable fuel savings, operational efficiency, and safety improvements at a price point and integration complexity that shortline operators can actually afford.

RailVision is currently raising a \$3M strategic round to execute its go-to-market plan, achieve a \$2.5M ARR target within 12 months, and position for an exit in the \$50M-\$100M range.

The single most important strategic insight: the only credible competitor in energy management (Wabtec Trip Optimizer) is entrenched in the Class I market, charges \$15,000-\$30,000 per locomotive as a capital expenditure, and has no documented shortline offering. RailVision can own this market before incumbents pivot.

Market Opportunity

North America operates an estimated 29,500 locomotives across freight, passenger, and transit categories:

- Union Pacific: ~7,300 locomotives
- BNSF Railway: ~6,000 locomotives
- Norfolk Southern: ~4,000 locomotives
- CSX Transportation: ~3,800 locomotives
- Canadian National (CN): ~2,000 locomotives (U.S.)
- Shortline Operators (combined): ~15,000 locomotives
- Passenger & Transit: ~1,200 locomotives

The shortline segment represents the single largest cohort -- approximately 15,000 locomotives -- and is predominantly comprised of Tier 0 (pre-2001) diesel units with minimal emission controls and no access to advanced optimization technology.

Addressable Market Calculation (Directional):

- Assumed active fleet: ~10,000 shortline locomotives
- Target license fee: \$300/locomotive/month
- Annual license value: \$3,600/locomotive/year
- Total ARR potential (full penetration): ~\$36M

Phased Market Development:

Phase 1 -- Shortline Segment (current focus)

RailVision -- Strategic Overview

Phase 2 -- Class I Railroads (scale phase)

Phase 3 -- Global Expansion

Fuel is one of the largest controllable operating costs for any railroad. Even low single-digit percentage fuel savings create material economic value at network scale, making ROI narratives highly credible for shortline operators with tight operating margins.

Competitive Landscape

The primary competitor in rail energy management is Wabtec Trip Optimizer -- an EPA-certified solution deployed on approximately 12,000 locomotives globally, exclusively serving Class I and major international railroads.

Wabtec Trip Optimizer -- Key Facts:

- Deployed on: ~12,000 locomotives (all North American Class I railroads)
- EPA-certified fuel savings: ~10% via optimized throttle and dynamic brake control
- Estimated per-locomotive cost: \$15,000-\$30,000 (capital expenditure model)
- Capabilities: Smart cruise control, air-brake advisement, LOCOTROL integration, PTC/ATP integration, Zero-to-Zero automation, Network Pacing
- Market focus: Class I railroads exclusively -- no documented shortline deployment

Wabtec's Separate Shortline Product:

Wabtec does offer a Transportation Management System (TMS) for shortlines -- managing movement of 9M+ annual carloads across 520 short lines -- but this is a logistics/dispatch tool, NOT a fuel optimization or energy management system.

Key Competitive Gaps Wabtec Cannot Easily Close:

1. Price -- \$15K-\$30K CapEx is prohibitive for shortline operators
2. Integration complexity -- requires PTC, LOCOTROL, and locomotive hardware compatibility
3. Locomotive specificity -- not designed for legacy Tier 0 fleets
4. Organizational focus -- Class I scale contracts dominate Wabtec strategic attention

RailVision strategic window: the period before any large incumbent pivots to serve the shortline segment with a purpose-built, affordable solution.

RailVision's Differentiation

RailVision is building the only fuel optimization and operational intelligence platform designed specifically for the shortline railroad market.

Core Differentiators:

RailVision -- Strategic Overview

1. Locomotive-Agnostic Architecture

RailVision technology is designed to work across locomotive types and tiers -- including legacy Tier 0 units that dominate the shortline fleet. No proprietary hardware integration is required, enabling rapid deployment without capital-intensive retrofits.

2. Tablet-Based, Low-Friction Deployment

The platform is delivered via a tablet interface, dramatically reducing deployment time and complexity. Shortline operators can onboard without specialized IT infrastructure or long integration cycles.

3. OpEx-Friendly Pricing Model

At ~\$300/month per locomotive, RailVision offers a subscription model that aligns with shortline budget cycles -- compared to Wabtec's \$15,000-\$30,000 upfront capital expenditure. The annual cost per locomotive (\$3,600) is 4-8x lower than the estimated annual cost of Trip Optimizer.

4. AI + Safety Fusion

RailVision combines AI-vision safety capabilities with fuel efficiency modules -- offering a bundled safety + eco-optimization system. This dual value proposition increases stickiness and justifies adoption even when fuel savings alone may be harder to quantify initially.

5. Shortline Operator Network

Founder Jamie brings a 20-year relationship network within the railroad community -- enabling trust-based pilot conversations and early customer access that a new market entrant cannot replicate.

Revenue Model & Financial Targets

Revenue Model:

- Primary: SaaS subscription license -- ~\$300/locomotive/month
- Model Type: OpEx (monthly subscription), not CapEx
- Billing Unit: Per active locomotive
- Target Customer: Shortline railroad operators

Current Strategic Fundraise: \$3M

Allocation:

- \$1.8M >> Technical team build-out and key hires
- \$1.2M >> Field rollout (GWRR pilot + initial scaling)

12-Month Financial Targets:

- ARR Target: \$2.5M
- Implied Active Locomotives (at \$300/month): ~694 locomotives
- Market penetration at target: ~7% of assumed 10,000 active shortline fleet

RailVision -- Strategic Overview

Exit Scenario:

- Target Exit Valuation: \$50M-\$100M
- Exit Multiple Implied: 20x-40x ARR (consistent with AI/SaaS/deep-tech valuations)
- Exit Strategy: Strategic acquisition by rail technology or industrial IoT acquirer (e.g., Wabtec, Alstom, Hitachi Rail)

Capital Efficiency Note:

The \$3M raise is structured to achieve ARR scale without excessive dilution -- a deliberate approach to preserve founder and early investor equity ahead of a potential strategic exit.

Technology Roadmap

RailVision technology roadmap is structured around three capability layers, built sequentially to maximize near-term commercial traction while creating long-term defensibility.

Layer 1 -- Core Platform (Current):

- AI-vision safety module (existing strength)
- Fuel-saving / eco-driving guidance (throttle, braking, horsepower management)
- Tablet-based operator interface
- Locomotive-agnostic data collection

Layer 2 -- Enhanced Intelligence (Near-Term):

- Integration with control systems (PTC, distributed power, LOCOTROL)
- Real-time fuel consumption telemetry and dashboarding
- Predictive maintenance signals from operational data
- Validated fuel savings reporting for ROI justification

Layer 3 -- Platform Scale (Medium-Term):

- Network-level optimization across multiple locomotives and corridors
- Emissions credit tracking and reporting (regulatory compliance layer)
- API integrations for railroad TMS and dispatch systems
- Expansion to Class I and international markets

Key Technology Decisions Pending:

- Extent of PTC/control system integration vs. standalone advisory model
- Hardware vs. pure software deployment architecture
- Partnership vs. in-house build for advanced automation modules
- AI model training data acquisition strategy (shortline-specific telemetry)

Fuel Savings Validation Approach

A credible, auditable fuel savings validation methodology is critical to RailVision commercial success --

RailVision -- Strategic Overview

particularly for shortline operators who need to justify subscription spend against measurable ROI.

Recommended Validation Partners:

- LB Foster (Rail Technologies) -- established rail supplier with proven friction management programs and performance validation frameworks
- First Analytics -- boutique analytics and AI consultancy specializing in statistical validation of railroad fuel-saving technologies (Rob Stevens, co-author of 'Analytical Approaches to Validating Railroad Fuel Saving Technology')

Core Validation Methodology:

1. Define KPIs upfront: fuel gallons per gross ton-mile (GTM), fuel per train-mile, throttle notch distribution
2. Establish baseline: 3-6 months pre-treatment data collection
3. Control design: treated vs. untreated locomotive groups where possible
4. Statistical methods: Difference-in-Differences (DiD), regression with covariates, Interrupted Time Series (ITS)
5. Uncertainty quantification: Bootstrap confidence intervals on all savings estimates
6. Business case translation: Convert % fuel saved to \$ saved at current diesel prices

Pilot Design Recommendation:

- Select 2-3 shortline corridors with varied operating characteristics
- Minimum 6-month baseline + 3-6 month treatment window
- Pre-register the statistical analysis plan before deployment

Key Risks to Manage:

- Confounding operational changes (crew changes, schedule shifts, locomotive swaps)
- Small sample sizes -- require longer time windows or multi-shortline pooled analysis
- Seasonal and weather effects -- must be modeled as covariates
- Fuel measurement accuracy -- reconcile fuel purchase records to individual locomotive assignments

Strategic Next Steps

Priority Actions (Next 90 Days):

1. Customer Discovery -- Shortline Operators

Conduct structured interviews with shortline railroad operators to validate:

- Perceived ROI for fuel optimization technology
- Alternatives currently in use or considered
- Real-world experience with RailVision to date
- Willingness to pay at \$300/month per locomotive
- Decision-making process and budget authority

RailVision -- Strategic Overview

2. Market Sizing -- Validation

Develop a rigorous, defensible market size for the shortline segment:

- Confirm Tier 0 locomotive count (~15,000 units cited)
- Identify the active, addressable subset
- Assess Trip Optimizer applicability gaps in the shortline tier

3. GWRR Pilot Execution

Execute the GWRR (Great Western Railway of Colorado) pilot as the marquee proof-of-concept:

- Establish baseline fuel consumption data
- Deploy RailVision platform
- Validate savings with pre-registered statistical methodology
- Document case study for investor and customer conversations

4. Partnership Conversations

Advance structured conversations with Wabtec, Alstom, and Hitachi Rail as both:

- Channel partners (distribution leverage in the shortline segment)
- Potential acquirers (exit path validation)

5. Class I Intelligence Gathering

Interview Class I customers to understand Trip Optimizer satisfaction levels, pricing structure, and gaps -- to identify future Class I entry vectors and validate Phase 2 of the market development plan.

Success Metric (12 Months): \$2.5M ARR -- validating product-market fit and positioning the company for a \$50M-\$100M strategic exit.